Alen Zimic Sheen

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EDUCATION

2019 - 2024
Doctor of Philosophy (PhD) in Computational Biology, Cornell University, Ithaca, NY
Funded by the SUNY Distinguished Fellowship - Dean Scholar

2019 - 2024
Bachelor of Science (B.Sc.) in Environmental Engineering, Universidad Peruana Cayetano Heredia (UPCH), Lima, Peru
GPA: 3.86; 2nd place in order of merit among the complete program of Environmental Engineering (16.9/20.00 in original scale)

RESEARCH EXPERIENCE

Laboratory Technician — **Aug 2025–Aug 2026**

Plant Science Department, Cornell University — *Ithaca, NY*
PI: Gaurav Moghe, PhD

- **Mechanistic discovery of plant metabolite functions from scientific literature**
  - Developing Python and LLM-assisted workflows to extract validated metabolite–gene/protein–trait relations from full-text plant biology articles.
  - Incorporating sentence-level provenance, experimental context, species, tissue, and stress-condition metadata to support evidence-grounded biological interpretation.
  - Building a knowledge resource for graph-based deep learning models aimed at predicting plant metabolite functions and stress-resilience mechanisms.

Visiting Student Researcher — **Jan 2025–May 2025**

Plant Gene Expression Center, UC Berkeley–USDA — *Albany, CA*
PI: Devin Coleman-Derr, PhD

- **Longitudinal RNA-seq of plant responses to synthetic microbial communities under drought**
  - Prepared plant microbox experiments for time-dependent evaluation of plant responses to synthetic community inoculation under drought stress.
  - Performed RNA extraction from soil, roots, shoots, and leaves.
  - Conducted phenotypic evaluation of roots, shoots, and leaves, including length, mass, and branching measurements.
- ***Streptomyces* diversity and function in sorghum roots under drought – Published in *PLOS Biology***
  - Conducted functional genomic analysis of 12 *Streptomyces* whole-genome sequences.
  - Analyzed genomic and phenotypic associations related to drought response and plant-growth effects.

Research Assistant
Jun 2021–Jan 2025

Bioinformatics and Molecular Biology Laboratory, UPCH — *Lima, Peru*

PIs: Mirko Zimic, PhD; Patricia Sheen, PhD

- **Immunoinformatics and antigen modeling – Published; collaboration with Dr. Robert Gilman, Cysticercosis Group in Peru.** Predicted and compared linear T-cell and conformational B-cell epitopes; performed global and local antigen alignments using LALIGN; and predicted MHC-I/MHC-II epitopes for Peruvian native population alleles.
- ***Taenia solium* adhesin-protein discovery.** Built adhesin-protein databases, conducted blastX searches against the *Taenia solium* genome and transcriptome, and reported candidate ORFs using high-scoring pair coordinates.
- **Lake Titicaca environmental metagenomics – Collaboration with Dr. Robert Gilman, Johns Hopkins University.** Trained in DNA preparation from water samples and processed metagenomic sequencing data using taxonomic classification pipelines.
- **Surface-display platform in *Salmonella enterica* Enteritidis.** Modeled truncated-INP fusion proteins for heterologous surface display and evaluated protein topology, structural quality, and alignments.

Bachelor's Thesis Researcher
Jan 2024–Jan 2025

Laboratory of Functional Genomics, UPCH — *Lima, Peru*

PI: Luis De-Stefano, PhD

- **Thesis:** "Effect of biol, a bovine biofertilizer, on taxonomic and functional changes in the rhizosphere microbiome of avocado trees in an agro-export farm in Chincha, Ica, Peru."
- Designed and implemented a pilot field study; performed DNA extraction, purification, library preparation, and Oxford Nanopore PromethION sequencing.
- Processed sequencing data using pipelines for taxonomic classification, assembly, functional annotation, and statistical analysis in R.

Research Assistant
Sep 2023–Jan 2025

Laboratory of Andean-Amazonian Chemistry of Life, UPCH — *Lima, Peru*

PI: Denis Castillo, PhD

- Developed alginate–xanthan gum hydrogels for controlled urea release in sustainable agriculture.
- Planned and conducted water-retention, biodegradability, and phytotoxicity assays; analyzed hydrogel physical and chemical properties using R and Python.

Trainee
Dec 2022–Mar 2023

Laboratory of Molecular Microbiology, UPCH — *Lima, Peru*

PI: Monica Pajuelo, PhD

- Trained in antibiogram evaluation of wastewater samples.
- Assisted in experimental design for metagenomic evaluation of Peruvian wastewater samples, with emphasis on resistome profiling.

PUBLICATIONS

Manuscript under review

A Zimic-Sheen; Moghe, G. AI-guided characterization of plant enzymes and metabolic pathways. *Phytochemistry Reviews* (2026). *In review.*

Peer-reviewed articles

Moghe, G.; **A Zimic-Sheen**; et al. Reimagining plant science training in the era of generative AI: A global perspective. *The Plant Cell* (2026). doi:10.1093/plcell/koag140.

Fonseca-Garcia, C.; Pettinga, D.; **A Zimic-Sheen**; et al. *Streptomyces*' role and hidden diversity in sorghum roots under drought. *PLOS Biology* (2025). doi:10.1371/journal.pbio.3003526.

Latorre, P.; **A Zimic-Sheen**; et al. Microfluidic chip printing protocol for chip characterization. *Progress in Additive Manufacturing* (2025). doi:10.1007/s40964-025-01391-z.

Monroy-Cruz, L. J.; **A Zimic-Sheen**; et al. Electrochemical determination of Pb(II) and Cu(II) on a FTO electrode modified with Ag nanoparticles stabilized with Nafion. *PLOS ONE* (2025). doi:10.1371/journal.pone.0320227.

Toribio, L.; Guzman, C.; Noazin, S.; **A Zimic-Sheen**; et al. Multiantigen print immunoassay (MAPIA) for the diagnosis of neurocysticercosis: A single-center diagnostic optimization and accuracy study in Lima, Peru. *Journal of Clinical Microbiology* (2022). doi:10.1128/jcm.00760-23.

Conference proceedings

Latorre-Quevedo, P. G.; **A Zimic-Sheen**; et al. An Open Hardware Bioprinter: A bioextruder with a peristaltic pump approach. *IEEE XXX International Conference on Electronics, Electrical Engineering and Computing* (INTERCON) (2023). doi:10.1109/INTERCON59652.2023.10326095.

Manuscripts in preparation

A Zimic-Sheen; Moghe, G. FuncMap workflow: Turning plant science literature into context-aware functional maps with local LLMs

A Zimic-Sheen; et al. Effects of inorganic fertilizers and biostimulants on taxonomic and functional shifts in the avocado rhizosphere microbiome. hdl:20.500.12866/17488.

Landa, M.; Balta, B.; **A Zimic-Sheen**; et al. Prediction and evaluation of adhesin-like proteins in the genome and transcriptome of *Taenia solium*.

FUNDING

Concytec–ProCIENCIA Applied Multidisciplinary Research Project Grant. **Aug 2025**
Designed and wrote the proposal "Identification, validation, and formulation of native microbial species with bioinoculant potential to increase water stress tolerance in avocado (*Persea americana* var. Hass) in the Lambayeque region." **Awarded: S/ 500,000 ($139,275) for three years.**

Secured funding for iGEM Grand Jamboree participation in Paris, France: **$10,000** from Agrícola Cerro Prieto as exclusive sponsor and **$8,000** from iGEM. **Oct 2024**

Supported preparation of an R01 NIH grant led by Dr. Robert Gilman, Johns Hopkins University, to evaluate soil microbiome composition under environmental stressors including floods and droughts. *Not awarded.* **Apr 2024**

Concytec–ProCIENCIA Applied Multidisciplinary Research Project Grant. Designed and wrote the proposal "Development of Smart Hydrogels with Controlled Release of Fertilizers and Nutrients for Organic Agriculture with Low Environmental Impact." *Not awarded.* **Feb 2024**

Funded with $2,500 by Maquinaria Agrícola San José to design and develop a metagenomic analysis evaluating liquid biofertilizer applications in agricultural soils. **Feb 2024**

MEDIA APPEARANCES

Speaker at TEDxLima on the genetic richness of Peruvian soil and its impact on society. **Oct 2024**

Featured in Agencia Andina, La República, Radio Nacional [radio interview], and the iGEM official newsletter for leading Paqta, the sole Peruvian team advancing to the iGEM Grand Jamboree Paris 2024, focused on mitigating soil degradation through synthetic biology. **Jul 2024**

HONORS AND AWARDS

SUNY Graduate Diversity Fellowship. Two years of funding for graduate studies. **May 2026**

NSF-GRFP Honorable Mention. **Apr 2026**

NSF-GRFP Honorable Mention. **Apr 2025**

Científico Latino Fellow. Selected as one of 100 scholars from 500+ applicants for personalized graduate admissions and scholarship-application guidance. **Aug 2024–Mar 2025**

iGEM 2024 Venture Creation Labs – Final Stage. Best Presentation Award and iGEM Grand Jamboree invitation as one of five global teams in the Food and Agriculture track. Developed a business model to improve *Persea americana* growth using synthetic biology to increase phytohormone production in rhizobacteria. **Mar 2024–Oct 2025**

Fully funded research expedition to Huaraz, Peru to investigate plant and microbiome capabilities for bioremediation of acid mine drainage and develop biotechnology research proposals. **Aug 2024**

HUC Social Ideas Challenge – Second Place. Designed a sustainable, low-cost hydrophobic bio-coating material for agriculture to reduce farmer exposure to agrochemicals. **Nov 2022–Mar 2023**

International Genetically Engineered Machine (iGEM) Design League – Team Leader; Second Place. Received university funding and a silver medal for developing a synthetic-biology biofungicide for Hass avocado trees. **Jan–Dec 2022**

POSTERS AND PRESENTATIONS

C-SPIRIT All-hands meeting, Japan - Riken Yokohama **Jul 2026**

Poster presentation: From Plant Science Literature to Contextualized Hypergraphs: Local LLMs for Gene–Compound–Trait Mechanism Discovery

24th Annual Microbiology Student Symposium, UC Berkeley. Poster presentation: *Streptomyces*' role and hidden diversity in sorghum root under drought – genomic analysis. **Apr 2025**

iGEM Grand Jamboree, Paris, France. Poster presentation. Sole Latin American team and one of five globally selected teams invited: genetically modifying plant-growth-promoting bacteria for agriculture. **Oct 2024**

Tech Startup Week, University of Lima. Oral presentation on behalf of Universidad Peruana Cayetano Heredia: bacterial consortia for improving agricultural efficiency as a metagenomic data-driven business model. **Jun 2024**

Science Poster Contest, UPCH Science Week. Oral presentation: evaluation of rhamnolipid overproduction in *Pseudomonas putida* for hydrocarbon bioremediation. **Nov 2023**

XXIV Jornadas Científicas "Dr. Abraham Vaisberg Wolach", Universidad Peruana Cayetano Heredia. Second Place. Oral presentation: development of electrochemical sensors for heavy-metal detection in aqueous solutions. **Oct 2023**

30th Peruvian Congress of Chemistry SILAE, Oral presentation: development of electrochemical sensors for heavy-metal detection in aqueous solutions. **Sep 2023**

PROFESSIONAL EXPERIENCE

External Consultant, Agrícola Cerro Prieto S.A.C. Led a $140,000 grant proposal: "Identification, validation and formulation of native microbial species with bioinoculant potential to increase tolerance to water stress in avocado (*Persea americana* Hass variety) in the Lambayeque-Peru region." **Sep 2024–Feb 2025**

Maquinaria Agrícola San José Designed and implemented soil microbiota analysis in avocado crop fields; evaluated fertigation regimes on rhizosphere taxonomy and genetic functionality across 72 Hass avocado trees using shotgun metagenomics. **Jan–Mar 2024**

Copacabana de Chincha S.A., Participated in anaerobic biodigester installation for wastewater treatment and reviewed company procedures for Peruvian environmental compliance and international agricultural export certification. **Jan–Apr 2022**

Farvet Laboratories, Laboratory Assistant Intern. Supported antibody-detection research using artificial intelligence for SARS-CoV-2 detection and gained experience developing molecular agglutination tests as immunology assays. **Jan–Apr 2022**

GITHUB REPOSITORIES

Transfer-Learning-of-ESM3-for-Enzyme-Substrate-Prediction. Transfer-learning project using frozen ESM3 protein foundation model representations to predict BAHD enzyme donor and acceptor substrate classes, with downstream classifier screening and interpretability analyses. **May 2026**

rotary-embeddings-demo. Interactive web demo for teaching rotary positional embeddings, including tokenization, positional encoding, RoPE rotation, theta-frequency effects, and cosine-similarity examples. **Apr 2026**

PupilTrackingDiagnosis. Computer vision program to track and analyze saccadic eye movements in response to a moving screen stimulus for early Parkinson's disease detection. **Jun 2024**

AutismSketchClassifier. Used ResNet-derived feature vectors and KNN classification to detect autism-associated features in children's sketches. **Apr–Jun 2024**

TEACHING EXPERIENCE

Research Experiences for Undergraduates (REU) mentor, Cornell University. **Jun 2026**

Plant Biochemistry master's student mentor, Cornell University. **Nov 2025–Present**

Leader, Journal Club of Genetic Engineering in Bioremediation, UPCH. Designed and led discussions on organic and heavy-metal bioremediation methods. **Aug–Dec 2023**

Leader, Journal Club of Applied Mycology, UPCH. Led discussions on recent applications of fungi in industry. **Feb–Jul 2023**

LEADERSHIP AND OUTREACH

Student co-lead, Cornell Initiative for Computational Innovation in Plant Sciences (I-CIPS). **Nov 2025–Present**

Member, Engineering Plan Committee, International Genetically Engineered Machine (iGEM) Foundation. Support plant-based synthetic biology projects from teams worldwide. **Oct 2024–Present**

Member, Board of Directors, Journal Club, UPCH. Managed logistics and communications. **Jan–Dec 2022**

Writer and later Head of Environmental Engineering, student magazine *The Novice Scientist*, Universidad Peruana Cayetano Heredia. Publications: Article 1, Article 2, Article 3, Article 4. **Jan–Dec 2021**

COURSES

Introduction to Deep Learning, graduate-level course, Cornell University. **Jan–May 2026**

Plant Biochemistry, graduate-level course, Cornell University. **Aug–Dec 2025**

Computer Vision, undergraduate elective in Informatics Engineering. **Mar–Jul 2024**

Bioinformatics I, master's-level course, Universidad Peruana Cayetano Heredia. **Sep–Oct 2023**

OTHER INTERESTS

First place, UPCH all-around foosball tournament, university-wide. **Jun 2024**

Theater and drama studies; member of university improv cast, with performances at UPCH Theater, Club de Teatro de Lima, San Martín de Porres, and UNI. **2015–Jan 2025**

Youth confirmation catechesis leader; mentored 50+ adolescents and organized community-service projects in nursing homes, orphanages, and low-income areas. **2018–2023**

First place, ADECOPA national basketball tournament with San Agustín School. **Jan–Aug 2018**